Customizing TeXmacs

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This tutorial shows how TeXmacs organizes its keyboard and its menus, and how you can add your own menus, keyboard shortcuts and icons. All the customizations go into one Scheme file of yours, which TeXmacs reads each time it starts:

`$TEXMACS_HOME_PATH/progs/my-init-texmacs.scm`

Here `$TEXMACS_HOME_PATH` is your personal TeXmacs folder: `~/.TeXmacs` on Linux and macOS, `%appdata%\TeXmacs` on Windows (TeXmacs creates its folders `progs` and `misc/pixmaps`). With the developer tool (Tools→Developer tool), the menu Developer→Open my-init-texmacs.scm opens this file for editing. After a change, restart TeXmacs to see its effect.

The complete file used for this tutorial, together with its icons, is `my-init-texmacs.scm` (the icons are in `images/icons`; copy them to `$TEXMACS_HOME_PATH/misc/pixmaps`, see section 5).

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1 The TeXmacs keyboard

TeXmacs uses four modifier keys, written as follows in the manual and in the Scheme files:

Modifier	Meaning	PC keyboard	Apple keyboard
`S-`	Shift	Shift	Shift
`C-`	Control	Ctrl	Control
`A-`	Alternate	Alt	Option
`M-`	Meta	Windows key	Command

Table 1. The modifier keys of TeXmacs.

The Hyper modifier `H-` of the old versions of TeXmacs, which one had to map onto the Caps-lock key, is gone: special symbols are now entered with the →| key (in a formula, P →| gives π, A →| gives α, < = gives $\leqslant$) or with the ⇧F5 prefix.

Which modifier does what depends on the *look and feel* chosen in Edit→Preferences→General: by default the one of your system. The shortcuts are therefore written in terms of three *prefixes*, which the look and feel translates into modifiers:

Prefix	Use	macOS	Windows, Linux
`std`	standard (copy, save)	`M-` (Command)	`C-` (Ctrl)
`cmd`	mode commands	`C-` (Control)	`A-` (Alt)
`altcmd`	compound commands	`M-C-` (Cmd+Ctrl)	`M-` (Windows key)

Table 2. The main prefixes of the keyboard shortcuts.

For instance, `std s` saves the document (Command-S on a Mac, Ctrl-S elsewhere), `math s` (that is `cmd s` in a formula) inserts a square root and `altcmd t` starts the table commands. On the Mac which produced this document they are shown as `⌘S`, `^S` and `⌘^T`. When the system takes a combination for itself, the `⎋` key gives the modifiers instead: on MACOS, `⎋` acts as `A-`, `⎋⎋` as `C-` and `⎋⎋⎋` as `M-`.

TEXMACS can also show a keyboard on the screen, which you can click (Developer→Show keyboard, with the developer tool):



Figure 1. The keyboard of TEXMACS on the screen.

2 How TEXMACS defines its own menus

The menus of TEXMACS are written in SCHEME, in the folder `$TEXMACS_PATH/progs` of the installation. The menu bar is defined in `progs/texmacs/menus/main-menu.scm` (abridged here):

```
(menu-bind texmacs-menu
  (=> "File" (link file-menu))
  (=> "Edit" (link edit-menu))
  (if (in-graphics?)
      (=> "Insert" (link graphics-insert-menu))
      (link texmacs-extra-menu)
      (=> "Focus" (link graphics-focus-menu)))
  (if (not (in-graphics?))
      (=> "Insert" (link insert-menu))
      (if (or (in-source?) (with-source-tool?))
          (=> "Source" (link source-menu)))
      ...
      (link plugin-menu)
      (link texmacs-extra-menu)
      (=> "Focus" (link focus-menu))
      (=> "Format" (link format-menu)))
  (=> "Document" (link document-menu))
  (if (project-attached?) (=> "Project" (link project-menu)))
  (=> "View" (link view-menu))
  (=> "Go" (link go-menu))
  (if (detailed-menus?) (=> "Tools" (link tools-menu)))
  ...
  (if (with-developer-tool?)
      (=> "Developer" (link developer-menu)))
  (=> "Help" (link help-menu)))
```

- The constructor `menu-bind` defines (or redefines) a menu.
- The constructor `link` inserts a menu defined elsewhere.

- `(=> "File" ...)` makes a pulldown menu in the menu bar.
- `if` shows entries only when a condition holds: the Source menu, for instance, only appears in source mode or with the source macros tool.

The File menu comes from `progs/texmacs/menus/file-menu.scm`:

```
(menu-bind file-menu
  ("New" (new-document))
  ("Load" (open-document))
  ("Revert" (revert-buffer))
  (-> "Recent"
      (link recent-file-menu)
      ...)
  ---
  ("Save" (save-buffer))
  ("Save as" (choose-file save-buffer-as "Save TeXmacs file" "texmacs"))
  ---
  (link print-menu)
  ---
  (-> "Import"
      (link import-import-menu)
      ...)
  (-> "Export"
      (link export-export-menu)
      ---
      ("Pdf" (choose-file wrapped-print-to-file "Save pdf file" "pdf"))
      ...)
  ---
  (if (window-per-buffer?)
      ("Close window" (close-document)))
  (if (not (window-per-buffer?))
      ("Close document" (close-document)))
  ("Close TeXmacs" (safely-quit-TeXmacs)))
```

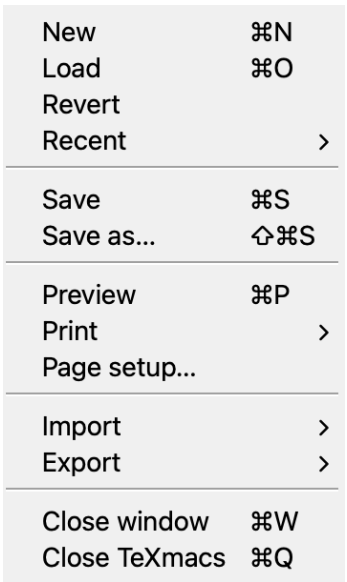


Figure 2. The File menu defined above.

An entry is a label followed by the SCHEME code to execute; `(-> "Import" ...)` makes a submenu (pullright) and `---` a separator. The shortcuts in the menus are found by TEXMACS itself. They are defined in the same way, with `kbd-map`, in files such as `progs/generic/generic-kbd.scm` and `progs/math/math-kbd.scm`:

```
(kbd-map
  (:profile std)
  ("std a" (select-all))
  ("std b" (toggle-bold))
  ("std c" (kbd-copy))
  ("std s" (save-buffer))
  ("std v" (kbd-paste))
  ...)

(kbd-map
  (:mode in-math?)
  ("math f" (make-fraction))
  ("math s" (make-sqrt))
  ("< =" "<leqslant>")
  ("- >" "<rightarrow>")
  ...)
```

and the prefixes of section 1 are defined in `progs/texmacs/keyboard/prefix-kbd.scm`:

```
(when (like-macos?)
  (kbd-wildcards pre
    ("std" "M-" #t) ; Command
    ("cmd" "C-") ; Control
    ("altcmd" "M-C-") ; Command+Control
    ...))

(when (like-windows?) ; idem for Gnome, KDE, Emacs
  (kbd-wildcards pre
    ("std" "C-" #t) ; Ctrl
    ("cmd" "A-") ; Alt
    ("altcmd" "M-") ; Windows key
    ...))
```

3 How to create your own menus

The menu bar contains an empty menu, `texmacs-extra-menu`, reserved for the user. Put the following lines in your `my-init-texmacs.scm`:

```
(menu-bind texmacs-extra-menu
  (former)
  (=> "AAA" (link aaa-menu))
  (=> "BBB" (link aaa-menu)))

(menu-bind aaa-menu
  ("AAA 111" (insert "Menu: aaa 111"))
  ("AAA 222" (insert "Menu: aaa 222"))
  ---
  (-> "AAA 333" (link file-menu))
  (-> "AAA 444" (link file-menu)))
```

- The constructor `=>` makes a pulldown menu (here two new entries AAA and BBB of the menu bar).
- The constructor `->` makes a pullright menu (a submenu): AAA→AAA 333 opens the File menu.
- The constructor `---` separates groups of entries.
- `(former)` keeps what other parts of TeXmacs (some plugins and styles, such as `literate`) put into the same menu.

After a restart of TeXmacs, the two menus appear between Insert and Focus; AAA→AAA 111 and AAA→AAA 222 insert their text in the document:

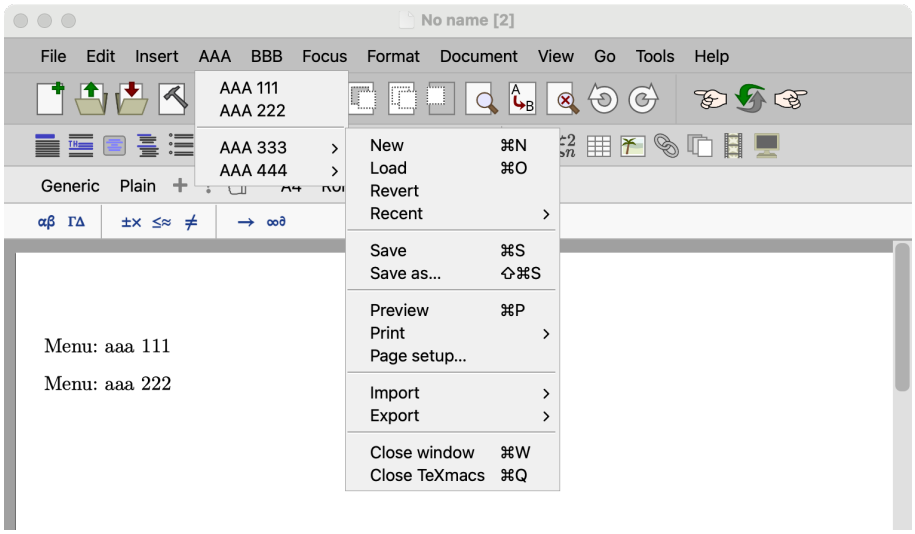


Figure 3. The new menu AAA, with its submenu AAA 333.

4 How to create your own keyboard shortcuts

Shortcuts are defined with `kbd-map`. The option `:mode` limits them to a mode: `in-text?`, `in-math?`, `in-prog?`... A key combination may be a single modified key (`"M-A-w"`) or a sequence of keys separated by spaces (`"e q u ."`): when you type the last key of a sequence, the characters typed before are removed and the action is done.

```
;; Keyboard shortcuts only valid in text mode

(kbd-map
  (:mode in-text?)
  ("M-A-w" (insert "Welcome to TeXmacs!"))
  ("e q u ." (insert-go-to "Eq. ()" '(5)) (make 'reference) (insert "equ:"))
  ("f i g ." (insert "Fig. ") (make 'reference) (insert "fig:"))
  ("t a b ." (insert "Table ") (make 'reference) (insert "tab:"))
  ("r e f ." (make 'cite) (insert "bib:")))

;; Keyboard shortcuts only valid in math mode

(kbd-map
  (:mode in-math?)
  ("g r a d ." (insert "<nabla>"))
  ("d i v ." (insert "<nabla><cdot>"))
  ("r o t ." (insert "<nabla><times>"))
  ("e q u ." (make 'label) (insert "equ:")))
```

Notice that the same sequence `e q u .` does different things in text and in formulas. Typing, in text mode, ⌘⌥W, and then `grad.`, `div.`, `rot.` in a formula, `equ.max` at the start of an equation and `equ.max`, `fig.lines`, `tab.data`, `ref.jackson` in the text gives:

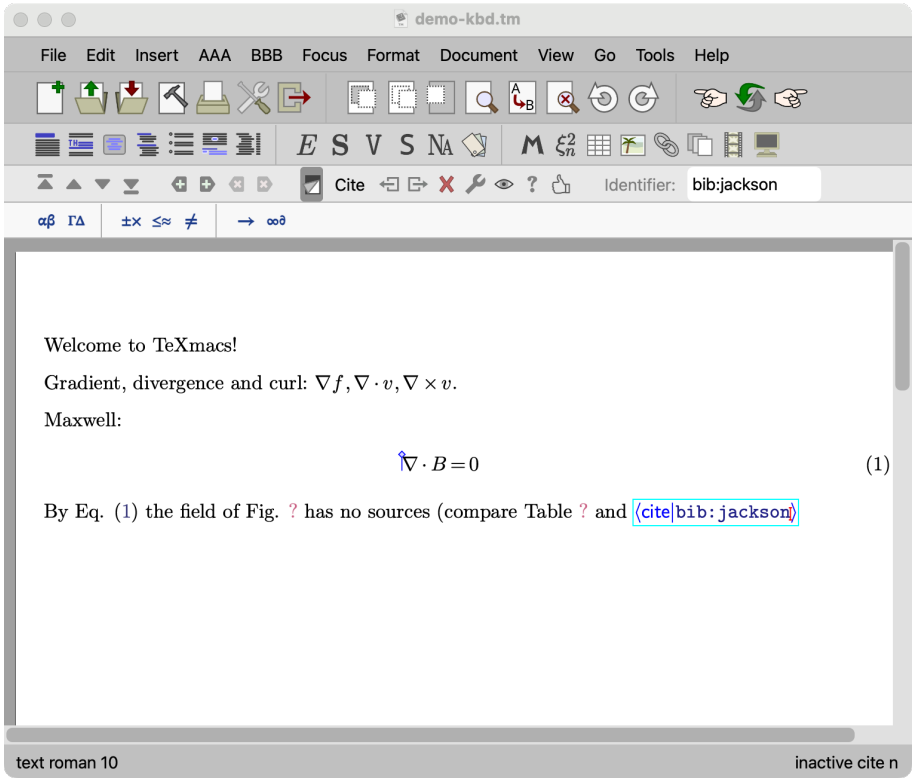


Figure 4. The new shortcuts at work: the citation is still being typed (the references to a figure and a table which do not exist are shown as ?).

The result of a shortcut can also be a string: `("p i ." "<pi>")` inserts π. A shortcut which already exists in TeXmacs for the same mode is replaced by yours (a shortcut of a more specific mode, for instance of formulas, still takes precedence in that mode).

5 How to create your own icon bar

Below the toolbars of TEXMACS there is room for a bar of your own icons, `texmacs-extra-icons`. Each icon opens a menu; here the menus are grids (`tile`) of mathematical symbols:

```
(menu-bind texmacs-extra-icons
  (former)
  (=> (balloon (icon "my_lowgre.png") "Lowercase Greek")
      (tile 6 (link my-lowercase-greek)))
  (=> (balloon (icon "my_uppgre.png") "Uppercase Greek")
      (tile 4 (link my-uppercase-greek)))
  |
  (=> (balloon (icon "my_binope.png") "Binary operations")
      (tile 5 (link my-binary-operations)))
  (=> (balloon (icon "my_binrel.png") "Binary relations")
      (tile 5 (link my-binary-relations)))
  (=> (balloon (icon "my_negrel.png") "Negated relations")
      (tile 5 (link my-negated-relations)))
  |
  (=> (balloon (icon "my_arrows.png") "Arrows")
      (tile 5 (link my-arrows)))
  (=> (balloon (icon "my_missym.png") "Miscellaneous symbols")
      (tile 5 (link my-miscellaneous-symbols))))
```

The original version of this tutorial had an eighth icon, for special delimiters ($\langle$, $\rangle$, $\lfloor$, $\rfloor$...); it is left as an exercise: add a menu `my-delimiters` and its icon in the same way.

with menus of symbols such as:

```
(menu-bind my-lowercase-greek
  (symbol "<alpha>") (symbol "<beta>") (symbol "<gamma>")
  (symbol "<delta>") (symbol "<epsilon>") (symbol "<varepsilon>")
  (symbol "<zeta>") (symbol "<eta>") (symbol "<theta>")
  (symbol "<vartheta>") (symbol "<iota>") (symbol "<kappa>")
  (symbol "<lambda>") (symbol "<mu>") (symbol "<nu>")
  (symbol "<xi>") (symbol "<pi>") (symbol "<varpi>")
  (symbol "<rho>") (symbol "<varrho>") (symbol "<sigma>")
  (symbol "<varsigma>") (symbol "<tau>") (symbol "<upsilon>")
  (symbol "<phi>") (symbol "<varphi>") (symbol "<chi>")
  (symbol "<psi>") (symbol "<omega>") (symbol "<varkappa>"))

(menu-bind my-uppercase-greek
  (symbol "<Gamma>") (symbol "<Delta>") (symbol "<Theta>")
  (symbol "<Lambda>") (symbol "<Xi>") (symbol "<Pi>")
  (symbol "<Sigma>") (symbol "<Upsilon>") (symbol "<Phi>")
  (symbol "<Psi>") (symbol "<Omega>") (symbol "<digamma>"))
```

- The constructor `|` separates groups of icons.
- `(balloon (icon "my_lowgre.png") "Lowercase Greek")` shows the icon, with a tooltip.
- `(tile 6 ...)` lays out the entries of a menu in a grid of 6 columns.
- `(symbol "<alpha>")` shows a symbol and inserts it in the document.

The icons are `.png`, `.svg` or `.xpm` files that you put in the folder

`$TEXMACS_HOME_PATH/misc/pixmaps`

For a sharp picture on high resolution screens, give the icon (20×20 pixels) a companion of twice the width and height (40×40) and the suffix `_x2`: `my_lowgre.png` and `my_lowgre_x2.png`. The icons of this tutorial are $\alpha\beta$ $\Gamma\Delta$ $\pm\times$ $\leq\approx$ $\neq$ $\rightarrow$ $\infty\partial$.

Finally, the bar of the user's icons must be made visible, with View→User provided icons:

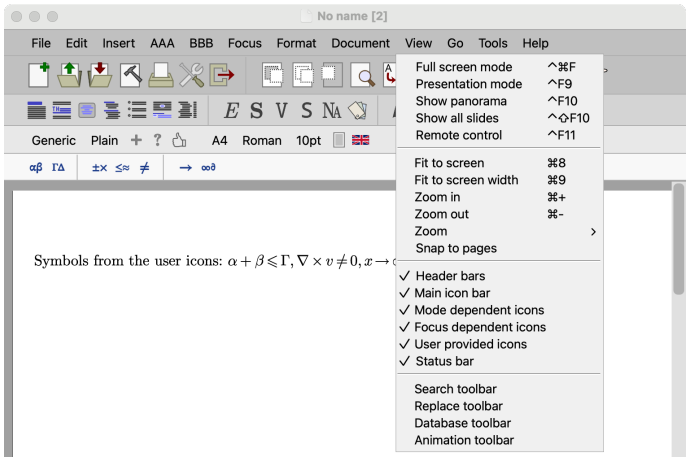


Figure 5. View→User provided icons.

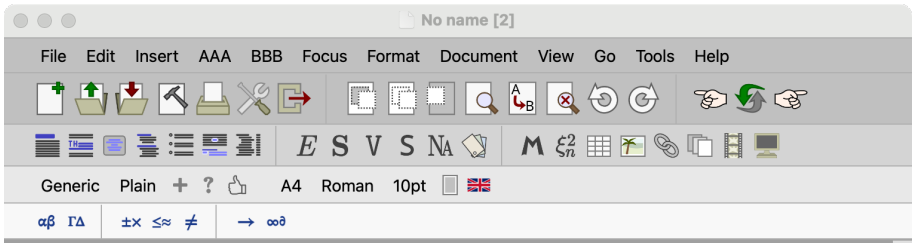


Figure 6. The bar of the user's icons, below the toolbars of TEXMACS.

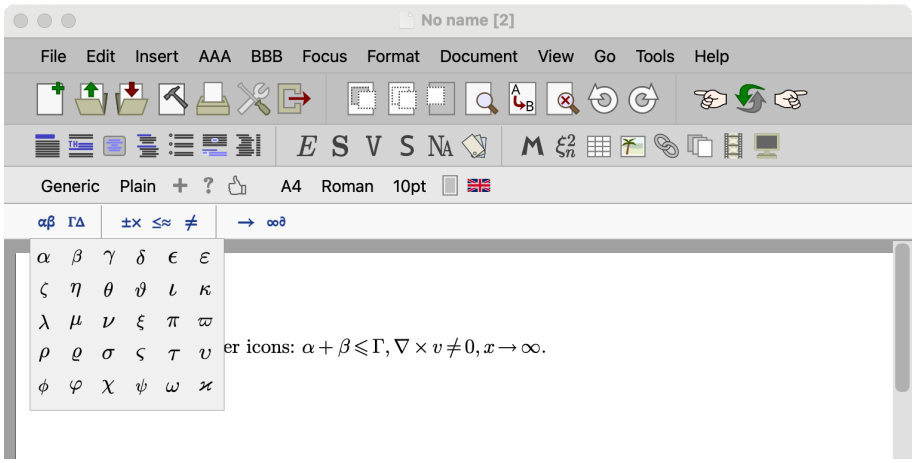


Figure 7. The menu of the first icon.

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